# Dongdong Yang


## Senior Switch Software Architect · Networking Systems & Linux Kernel

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## PROFILE

Software architect and engineering leader with 20 years of experience building networking products and high-performance systems. Deep background in Ethernet switching, IP/MPLS routing, Linux kernel and drivers, DPDK packet processing, embedded platforms, and control/data-plane architecture. Former technical lead for Huawei data-center switching features and platform manager for Juniper SRX. Hands-on coder across C/C++, Go, Rust, Zig, and Python; inventor on 10 networking and security patents; founder of a 12-person engineering organization serving 100+ enterprise customers.

**Direct fit:** switch software architecture · Linux networking · embedded systems · feature prototyping · cross-functional technical leadership · patents, open source, and technical publishing

## CORE EXPERTISE

**Switching & Routing**
Ethernet · TRILL · TCP/IP · IP/MPLS · BGP · OSPF · IS-IS · ARP · IP FRR · NSR · QoS · traffic shaping · data-center topologies

**Systems & Data Plane**
Linux kernel/network stack · BSP · device drivers · DPDK · SR-IOV · VirtIO · interrupts · memory management · performance profiling

**Architecture & Platforms**
Control/data-plane design · embedded software · NOS/platform integration · x86/MIPS/ARM · KVM/QEMU · high availability · validation strategy

**Leadership & Coding**
Architecture reviews · technical roadmaps · PoCs · customer engagement · cross-site teams · C/C++ · Go · Rust · Zig · Python

## PROFESSIONAL EXPERIENCE

**Founding Engineer / Architect** — Apr 2018 — Present

**Servicewall.ai · Cloud-Native Network, API Security & AI Applications**

- Defined the end-to-end architecture of a distributed HTTP proxy and security gateway processing hundreds of millions of requests per day; translated product and customer needs into technical requirements, designs, and delivery plans.
- Built the company and product from 0 to 1, leading 12 engineers across platform, data, security, and application layers; established design/code reviews, CI/CD, test strategy, observability, and operational ownership.
- Integrated an eBPF/XDP-accelerated data plane into the traffic-processing platform, moving filtering and early processing into the Linux kernel ingress path to improve throughput.
- Built and optimized the end-to-end pipeline from packet ingress and security inspection to streaming and analytics, sustaining 50,000 QPS with Go, ClickHouse, Kafka, Redis, and Spark.
- Built AI-agent applications for security operations and data analysis, connecting LLM-driven workflows with traffic and security data pipelines; led customer-facing PoCs and converted emerging techniques into deployable products.
- Authored/co-authored five recent patents covering distributed traffic analysis, API anomaly detection, AI workload protection, and zero-trust access.

**Software Engineering Manager, Platform** — Apr 2013 — Apr 2018

**Juniper Networks · SRX Platform Team**

- Led SRX/NGSRX platform software development across BSP, Linux kernel, drivers, control-plane infrastructure, software data plane, virtualization, and high-performance forwarding on x86 and MIPS.
- Supported Broadcom-based NGSRX bring-up and MIPS Broadcom SDK adaptation, primarily owning platform integration, control-plane infrastructure, software FIB management, and the JEXEC forwarding engine.
- Designed a multi-table FIB programming pipeline that encoded Routing Entry and Next-hop structures as fixed-format forwarding instructions and downloaded them to JEXEC, a software forwarding engine and forwarding-chip simulator.
- Developed an FPGA driver for the NGSRX SPC service-processing card, integrating a dual-socket x86 platform with FPGA acceleration for firewall-service offload.
- Architected Daybreak, separating a VM-based control plane from a DPDK-accelerated data plane; delivered SRX1500 with KVM/QEMU, SR-IOV, and VirtIO for near-line-rate packet processing.
- Coordinated engineering across China, India, and the US; progressed from MTS4 to Staff Engineer and Manager.

## PROFESSIONAL EXPERIENCE — EARLIER

**Senior Software Engineer / Technical Lead** Jul 2006 — Mar 2013

**Huawei · Network Systems Platform, Beijing R&D**

- Served four years as technical lead for routing and Ethernet switching software, owning architecture, implementation, integration, and issue resolution across approximately ten product projects.
- Chief designer and hands-on developer of TRILL, a key multi-path Layer-2 data-center switching feature for the CloudEngine product line; received the 2012 Beijing R&D Design Award.
- Designed and implemented IP Fast Reroute and led an NSR prototype; optimized routing convergence to rank first in multiple competitive benchmarks.
- Worked across protocol, platform, forwarding, hardware, and test teams to map feature requirements to software modules and platform capabilities.
- Authored/co-authored five networking patents covering TRILL table programming, route backup, ARP entry configuration, multicast path switching, and routing-table synchronization.

## EDUCATION

**B.Eng., Computer Science & Technology · Shenyang University of Technology** 2002 — 2006

Top 10% · Three scholarships · ACM/ICPC, Topcoder, and Baidu Star competitor (National Top 300)

## SELECTED PRODUCTS & ARCHITECTURE IMPACT

- **Huawei CloudEngine / TRILL** — multi-path Layer-2 data-center switching feature; architecture, protocol implementation, table programming, convergence, and platform integration.
- **Huawei NetEngine 5000E** — carrier-class cluster router; routing protocols, fast convergence, resiliency, and control/forwarding-plane collaboration.
- **Juniper SRX300 / SRX500 / SRX1500** — embedded platform stack from BSP and kernel drivers to DPDK-accelerated packet processing.
- **Juniper Junos OS** — kernel and data-plane development, virtualization, high availability, IP/MPLS forwarding, QoS, and performance tuning.
- **Servicewall Security Gateway** — cloud-native distributed proxy and deep traffic-analysis platform operating at enterprise scale.

## OPEN SOURCE, CODING & PROOFS OF CONCEPT

**wFilter** · C / DPDK
High-performance web keyword firewall using DPDK and Aho–Corasick matching—an end-to-end networking PoC spanning packet processing, algorithms, and application control.

**ds4-ascend** · C / AscendC
Ported antirez's DS4 inference engine to Huawei Ascend NPU, integrating CANN runtime, custom operators, GGUF loading, and expert/weight caching.

**ClickFS** · Rust / FUSE
Read-only POSIX filesystem for browsing ClickHouse with standard shell tools; streaming data access and nine published releases.

**ZigHouse** · Zig
Columnar OLAP engine compatible with ClickHouse wire and MergeTree formats; implements SQL parsing, IR execution, SIMD, and parallel scheduling. A specialized ClickBench build completed all 43 queries and ranked first.

**llama2.zig** · Zig
Dependency-free Llama 2 inference implementation with tensor operations, tokenizer, Transformer forward pass, and portable systems-level code.

**go-yyjson** · Go / C
High-performance Go binding for yyjson with zero-copy views and streaming/DOM APIs, demonstrating low-level interoperability and performance engineering.

## PATENTS & TECHNICAL COMMUNICATION

- **10 issued/published patents** across switching, routing resilience, distributed traffic analysis, API security, and AI workload protection.
- Representative: TRILL table programming and controller synchronization (CN103414644A).
- Representative: route backup and rapid recovery / IP FRR (CN103179032A).
- Representative: ARP entry configuration (WO2012130083A1).

- **donge.org** — long-running technical blog focused on networking, AI, and low-level systems.
- Publishes design notes, implementation walkthroughs, benchmarks, reverse engineering, and lessons learned from open-source projects.
- Public portfolio includes 80 GitHub repositories spanning networking, kernels/systems, databases, AI inference, and developer tooling.